Astronomy Labs

Alveary

Astronomy Labs

GRADE(S): 11-12

About the Course

Labs/fieldwork are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Astronomy

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
11-12	Astronomy Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Astronomy				
Astronomy Lessons Nature Walks & Scouting: Grades 9-12	Astronomy Lessons	Astronomy Lessons	Astronomy Lessons Nature Notebook: Grades 9-12	Astronomy Lessons Astronomy Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

(No Subject Supplies Assigned)

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

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How To Approach Labs

Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

Astronomy Labs

Click THIS text or scan the QR code for links.

Term 1

WEEK 1

⬚ **60m Astronomy Labs - Lesson 1**

Materials: journal, star finder

➜ OBSERVE
Simply notice on this first night of observing. What are your first impressions? Or are you able to relocate any objects previously found? Do you notice anything new based on your use of the star finder or Quick Links? Keep a record of what you observe.

WEEK 2

⬚ **60m Astronomy Labs - Lesson 2**

Materials: journal, star finder

➜ OBSERVE
Notice any objects that are or are becoming familiar. In what directions do they currently lie? Using your hands, measure their locations in the sky. Do these locations change during the evening? Keep a record of what you observe.

★ STUDENT & TEACHER TIP
Note that next week's fieldwork should be done in one or more daytime sessions!

WEEK 3

⬚ **60m Astronomy Labs - Lesson 3**

Materials: journal, binoculars, paper, tripod/stand for binoculars

➜ OBSERVE
Use your hands to measure locations and your binocular set-up to observe the sun's projection. DO NOT LOOK DIRECTLY AT THE SUN! Keep a record of what you observe.

WEEK 4

⬚ **60m Astronomy Labs - Lesson 4**

Materials: journal, star finder, binoculars/telescope

➜ OBSERVE
Use your hands to measure locations of the moon and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope to view the moon up close. Keep a record of what you observe.

WEEK 5

⬚ **60m Astronomy Labs - Lesson 5**

Materials: journal, star finder, binoculars/telescope

➜ OBSERVE
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 6

⬚ **60m Astronomy Labs - Lesson 6**

Materials: journal, star finder, binoculars/telescope

➜ OBSERVE
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

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Term 1

WEEK 7 ⬚ **60m Astronomy Labs - Lesson 7**

Materials: journal, star finder, binoculars/telescope

→ OBSERVE
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 8 ⬚ **60m Astronomy Labs - Lesson 8**

Materials: journal, star finder, binoculars/telescope

→ OBSERVE
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 9 ⬚ **60m Astronomy Labs - Lesson 9**

Materials: journal, star finder, binoculars/telescope

→ OBSERVE
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 10 ⬚ **60m Astronomy Labs - Lesson 10**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of Ursa Major, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 11 ⬚ **60m Astronomy Labs - Lesson 11**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of Ursa Major and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 12 ⬚ **60m Astronomy Labs - Lesson 12**
Exam Week

→ EXAMS
Answer question(s) related to the course.

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Term 2

WEEK 1 ⬚ **60m Astronomy Labs - Lesson 13**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of Ursa Major, Ursa Minor, and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Has the location of Ursa Major changed from your last viewing? Do the locations change during your session? Keep a record of what you observe.

WEEK 2 ⬚ **60m Astronomy Labs - Lesson 14**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 3 ⬚ **60m Astronomy Labs - Lesson 15**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 4 ⬚ **60m Astronomy Labs - Lesson 16**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 5 ⬚ **60m Astronomy Labs - Lesson 17**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 6 ⬚ **60m Astronomy Labs - Lesson 18**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 7 ⬚ **60m Astronomy Labs - Lesson 19**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated

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stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 8

- [ ] **60m Astronomy Labs - Lesson 20**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 9

- [ ] **60m Astronomy Labs - Lesson 21**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 10

- [ ] **60m Astronomy Labs - Lesson 22**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 11

- [ ] **60m Astronomy Labs - Lesson 23**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 12

- [ ] **60m Astronomy Labs - Lesson 24**

Exam Week

→ EXAMS
Answer question(s) related to the course.

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WEEK 1 ⬚ **60m Astronomy Labs - Lesson 25**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 2 ⬚ **60m Astronomy Labs - Lesson 26**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 3 ⬚ **60m Astronomy Labs - Lesson 27**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 4 ⬚ **60m Astronomy Labs - Lesson 28**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 5 ⬚ **60m Astronomy Labs - Lesson 29**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 6 ⬚ **60m Astronomy Labs - Lesson 30**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 7 ⬚ **60m Astronomy Labs - Lesson 31**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

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WEEK 8

☐ **60m Astronomy Labs - Lesson 32**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 9

☐ **60m Astronomy Labs - Lesson 33**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 10

☐ **60m Astronomy Labs - Lesson 34**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 11

☐ **60m Astronomy Labs - Lesson 35**

Materials: journal, star finder

→ OBSERVE
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 12

☐ **60m Astronomy Labs - Lesson 36**
Exam Week

→ EXAMS
Answer question(s) related to the course.